Exercise 8

On convexity of curves and inflection points

From the school book | Understand | Apply | Higher Order Thinking Skills

1 **Determine the intervals of convexity upwards , downwards and the inflection points (If exist) for each of the following :**

(1) $f(x) = (x-1)^2$

(2) $f(x) = 4 - 6x - 3x^2$

(3) *(1st Session 2001)* $y = x(x^2-1)$

(4) $f(x) = x^3 - 6x^2 + 9$

(5) $f(x) = 15x + 6x^2 - x^3$

(6) $f(x) = (x-2)^3 - 1$

(7) $f(x) = (x-1)^4 + 3$

(8) $f(x) = (x-1)^2(x-2)$

(9) $f(x) = (x^2-4)^2$

(10) $f(x) = 4 - x - 3x^4$

(11) $y = x^4 - 4x^3 + 6x^2 + 5$

(12) *(2nd Session 2007)* $y = x^4 - 6x^2$

(13) $f(x) = 6x^3 - x^4$

(14) $f(x) = x^4 - 8x^2 + 16$

(15) *(2nd Session 2002)* $y = -\frac{1}{4}x^4 + x^2$

(16) $f(x) = -3x^5 + 5x^3$

2 **Determine the intervals of convexity upwards , downwards and the inflection points (If exist) :**

(1) $f(x) = x^2 - \frac{8}{x}$

(2) $f(x) = \frac{6}{x^2+3}$

(3) *(1st Session 2012)* $f(x) = \frac{x^2+9}{x}$

(4) $f(x) = \frac{x^2-1}{2x-1}$

(5) *(2nd Session 2012)* $f(x) = \frac{x^2+1}{x^2+3}$

(6) $f(x) = \sqrt[3]{x}$

(7) $f(x) = x^{\frac{2}{3}}$

(8) $f(x) = \sqrt[3]{(x-4)^2}$

(9) $f(x) = \sqrt[5]{(x-3)^3}$

(10) $f(x) = |x^3 - 1|$

3 **Choose the correct answer from the given ones :**

(1) If the curve of the function f is convex downwards in an interval , then ..... in this interval.

(a) $f'(x) > 0$ (b) $f'(x) < 0$ (c) $f''(x) > 0$ (d) $f''(x) < 0$

(2) The curve of the function f is convex upwards in an interval , if ..... in this interval.

(a) $f'(x) > 0$ (b) $f'(x) < 0$ (c) $f''(x) > 0$ (d) $f''(x) < 0$

(3) The function $f : f(x) = x^3 - 3x - 1$ has an inflection point at .....

(a) $(0 , 1)$ (b) $(0 , -1)$ (c) $(1 , 0)$ (d) $(-1 , 0)$

(4) *(1st Session 2022)* If the curve of the function $f : f(x) = a x^3 - 2x^2 + 4$ has an inflection point at $x = \frac{1}{3}$, then $\frac{f(1)}{\grave{f}(1)} =$ .....

(a) -2 (b) -1 (c) 1 (d) 2

(5) *(2nd Session 2018)* If the curve $y = (2x - a)^3 + 4$ has an inflection point at $x = 5$, then a = ...

(a) 2 (b) 4 (c) 5 (d) 10

(6) *(Speciman 2025)* If the curve $y = x^3 + a x^2 + b x$ has an inflection point at $(3 , -9)$, then a + b = ...

(a) 6 (b) -9 (c) 15 (d) 24

(7) The curve of the function f is convex downwards on $\mathbb{R}$ if $f(x)$ equals .....

(a) $3 - x^2$ (b) $3 - x^3$ (c) $3 - x^4$ (d) $3 + x^4$

(8) The curve of the function $f : f(x) = x^3 - 3x^2 + 2$ is convex upwards when $x \in$ .....

(a) $]-\infty , 0[$ (b) $]-\infty , 1[$ (c) $]1 , 3[$ (d) $]1 , \infty[$

(9) *(2nd Session 2022)* If $f : f(x) = \frac{x+1}{x}$, then the curve of the function f is convex upward on the interval .....

(a) $]0 , \infty[$ (b) $]-\infty , 3[$ (c) $]-1 , \infty[$ (d) $]-\infty , 0[$

(10) If $f(x) = \sqrt[3]{x-2}$ and the function has an inflection point at $(2 , 0)$, then $\grave{f}(2) =$ .....

(a) undefined. (b) zero (c) 1 (d) $-\frac{2}{9}$

(11) If the function $f : f(x) = x^4$, then the point $(0 , 0)$ is .....

(a) local minimum. (b) local maximum.

(c) inflection point. (d) a and c together.

(12) If the curve of the function f lies above all the tangents which drawn from all the point of the curve , then the curve of the function is .....

(a) convex upward. (b) decreasing.

(c) increasing. (d) convex downward.

(13) If f is a function of fourth degree , then the greatest possible number of the inflection points equals .....

(a) 2 (b) 1 (c) 3 (d) 4

(14) If the curve of the function $f : f(x) = a x^2 + \sin x$ where $a \in \mathbb{R}$ has an inflection point at $x = \frac{\pi}{6}$, then a = ...

(a) $-\frac{1}{4}$ (b) $\frac{1}{4}$ (c) $\frac{1}{2}$ (d) $\frac{\sqrt{3}}{4}$

(15) *(2nd Session 2018)* The curve of the function $f : f(x) = (x - 2)\, e^x$ is convex downwards on the interval .....

(a) $]-\infty , \infty[$ (b) $]-1 , 2[$ (c) $]-1 , \infty[$ (d) $]0 , \infty[$

(16) *(1st Session 2023)* If the curve of the function $y = a\, e^{\frac{b}{x}}$ has an inflection point at $(1 , 1)$ where a , b are constants , then $a\, b \in$ .....

(a) $\{0 , 2e^2\}$ (b) $\{-2e^2\}$ (c) $\{2e^2\}$ (d) $\{0\}$

(17) The curve of the function $f : f(x) = \begin{cases} x^3 + 2 & , x > 0 \\ 3 - 2x^2 & , x < 0 \end{cases}$ is convex upwards when $x \in$ .....

(a) $]0 , \infty[$ (b) $]-\infty , 0[$ (c) $\mathbb{R}$ (d) $\mathbb{R}^*$

(18) If the point $(1 , 2)$ is an inflection point of the curve of the function f where $\grave{f}(x) = 4x^3 - k x^2$, then k = ...

(a) 24 (b) 6 (c) 4 (d) 12

(19) The critical point of the function $f : f(x) = x^3 + 60$ is .....

(a) an inflection point. (b) a local maximum.

(c) a local minimum. (d) both (a) and (c)

(20) If f , g are two differentiable twice functions on $\mathbb{R}$ and $\acute{f}(x) < \acute{g}(x)$ for all values of x and $h(x) = f(x) - g(x)$, then h .....

(a) increases on its domain. (b) decreases on its domain.

(c) has a curve convex downwards. (d) has a curve convex upwards.

(21) If $\grave{f}(x) = (x - m)(x - k)$ where $m < k$, then .....

(a) $\grave{f}(m) = \grave{f}(k)$ (b) $\acute{f}(m) < \acute{f}(k)$

(c) The function f has an inflection point at $x = \frac{m + k}{2}$

(d) all the previous.

(22) If $y = a x + b$ is a tangent to the curve of the function f at any point on it , and $f(x) \leq a x + b$, then the curve of $f(x)$ ..... for all values of x

(a) increases (b) decreases

(c) convex downwards (d) convex upwards

(23) If $f''(x) = (x-3)(x+2)$, then the curve of the function f is convex upwards in the interval .....

(a) $]-\infty , -2[$ (b) $]-2 , 3[$ (c) $]3 , \infty[$ (d) $]-\infty , 3[$

(24) *(1st Session 2018)* The function $f : \mathbb{R} \longrightarrow \mathbb{R}$ such that $f'(x) = -2x + 6$, then all of the following statements are correct except .....

(a) the curve of the function f convex upwards in the interval $]-\infty , \infty[$

(b) the function f has a local minimum value at $x = 3$

(c) the curve of the function f has no inflection points.

(d) $f(x)$ is decreasing in the interval $]3 , \infty[$

(25) The curve of the function $f : f(x) = 3x^2 \ln 2x$ is convex upward and convex downward on the two intervals ..... respectively.

(a) $]0 , \frac{1}{2\sqrt{e^3}}[,]\frac{1}{2\sqrt{e^3}} , \infty[$ (b) $]\frac{1}{2\sqrt{e^3}} , \infty[,]0 , \frac{1}{2\sqrt{e^3}}[$

(c) $]0 , \frac{1}{2\sqrt{e}}[,]\frac{1}{2\sqrt{e}} , \infty[$ (d) $]0 , 2\sqrt{e}[,]2\sqrt{e} , \infty[$

(26) *(1st Session 2021)* f , g are polynomial functions , $f(x) = cx^2 + g(x)$, $g(1) = k$ and $g''(1) = 6$ where c , k are constants. If $(1 , 5)$ in an inflection point to the curve of f , then $k - c =$ .....

(a) -11 (b) 5 (c) 11 (d) -5

(27) *(Trial 2021)* If f is differentiable twice function in the interval $[-1 , 1]$ where $f'(x)$ is increasing on $]-1 , 0[$ and $f'(x)$ is decreasing on $]0 , 1[$, then the statement which is certainly correct from the following is .....

(a) $f(0)$ is a local maximum value of the function.

(b) the point $(0 , f(0))$ is an inflection point of the function.

(c) the function f is increasing on $]0 , 1[$

(d) the function f is decreasing on $]0 , 1[$

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4 If the point $(1 , 12)$ is the inflection point to the curve of the function f where $f(x) = ax^3 + bx^2$ Find the real values of a and b « $-6 , 18$ »

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5 *(2nd Session 2014) , (1st Session 2019)* Find the values of a and b such that curve $y = x^3 + ax^2 + bx$ has an inflection point at $(3 , -9)$ « $-9 , 15$ »

6 Find the inflection points of the curve of the function $f : f(x) = 2x + \sin x$ on the interval $]\frac{-\pi}{2}, \frac{\pi}{2}[$

« (0 , 0) »

7 📖 Prove that the measure of the angle of inclination of the tangent at the inflection point to the curve of the function f where $f(x) = \frac{x}{1-x^2}$ equals $\frac{\pi}{4}$

8 📖 If the curve of the function f where $f(x) = x(x-3)^2$ has a local maximum value at x_1 and a local minimum value at x_2 , prove that x – coordinate of the inflection point $= \frac{x_1 + x_2}{2}$

9 If $\hat{f}(x) = a x + b$ where $a < 0$

(1) Is the curve of $f(x)$ has a local maximum or minimum value ?

(2) Is the curve of $f(x)$ has an inflection point or not ?

10 Find a , b , c and d such that the curve $y = a x^3 + b x^2 + c x + d$ touches the x-axis at the point (2 , 0) and has an inflection point at (0 , 4)

« $\frac{1}{4}$, 0 , – 3 , 4 »

11 If the curve of the function f where $f(x) = a x^3 + b x^2 + c x + d$ has a local maximum value at (2 , 4) and an inflection point at (1 , 2)

Find the equation of the curve.

« $f(x) = -x^3 + 3x^2$ »

12 Find a function of the third degree if its curve passes through (1 , 0) , the point (2 , 1) is an inflection point and the equation of the tangent at the inflection point is $3x + 2y = 8$

« $f(x) = \frac{5}{2}x^3 - 15x^2 + \frac{57}{2}x - 16$ »

13 If the function $f(x) = a x^3 + b x^2 + 9x + 1$ has two critical points at $x = 1$, $x = 3$, then find a and b determine the convexity of the curve and find the points of inflection if they exist.

« 1 , – 6 , (2 , 3) is an inflection point »

14 If f is a differentiable function on $\mathbb{R}$, $f(x) = \begin{cases} 2x^2 + a x + b & , \text{ when } x \geq 1 \\ 3x - x^2 & , \text{ when } x < 1 \end{cases}$

(1) Find the values of the two constants a and b

(2) Determine the interval of being convex upwards and downwards and the inflection points if found.

« – 3 , 3 , (1 , 2) »

15 **Determine the convexity intervals of the curve of f upwards and downwards and find the inflection points and the equation of the tangent to each of the following function at this points :**

(1) $f(x) = \begin{cases} (x+3)^2 & , \ x < -1 \\ 3x^2 - x^3 & , \ x \geq -1 \end{cases}$ | (2) $f(x) = \begin{cases} x^2 - 4 & , \ x < -2 \\ x^3 - 3x + 2 & , \ x \geq -2 \end{cases}$

(3) $f(x) = \begin{cases} x^3 - 3x & , \ x < 0 \\ 4x - x^2 & , \ x \geq 0 \end{cases}$

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16 **By using the opposite table where the domain of the function f is $\mathbb{R}$ and f has no other critical or inflection points except those showing in the table choose the correct answer from the given ones :**

x	-1	1	2	3	4
$\dot{f}(x)$	24	0	-3	0	9
$\ddot{f}(x)$	-18	-6	0	6	12

(1) $f(x)$ has a local maximum value at $x =$ .....

(a) -1 (b) 1 (c) 2 (d) 3

(2) $f(x)$ has a local minimum value at $x =$ .....

(a) 1 (b) 2 (c) 3 (d) 4

(3) $f(x)$ is decreasing at $x \in$ .....

(a) $]-\infty , 1[$ (b) $]1 , 3[$ (c) $]3 , \infty[$ (d) $\mathbb{R}$

(4) The curve $f(x)$ is convex upwards when $x \in$ .....

(a) $]-\infty , -1[$ (b) $]-\infty , 2[$ (c) $]3 , \infty[$ (d) $]1 , \infty[$

(5) The curve $f(x)$ has an inflection point at $x =$ .....

(a) -1 (b) 1 (c) 2 (d) 3

(6) If $g(x) = f(x) + 5$, then $g(x)$ has a local maximum value at $x =$ .....

(a) -1 (b) 1 (c) 2 (d) 3

Higher skills

17 **Choose the correct answer from those given :**

(1) If the curve of a polynomial function has one inflection point , then the greatest number of points of intersection of the curve of the function and any straight line equals .....

(a) 1 (b) 2 (c) 3 (d) 4

(2) If $f(X) = (a-2)X^2 + 3X - 5$, $X \in \mathbb{R}$, then the curve of the function is concave downward when .....

(a) $a > 2$ (b) $a < 2$ (c) $a = 2$ (d) $a = 0$

(3) If the function f : $f(X) = g(X) - r(X)$ where $\grave{g}(2) = \grave{r}(2)$, $\acute{g}(2) > \acute{r}(2)$ at $X = 2$ the function f has .....

(a) a local minimum value. (b) a local maximum value.

(c) an inflection point. (d) an absolute maximum value.

(4) If f is a function such that $\lim_{h \to 0} \frac{\grave{f}(X+h) - \grave{f}(X)}{h} > 0$ for all values of $X \in \mathbb{R}$, then the curve of the function f is .....

(a) convex upward. (b) increasing.

(c) decreasing. (d) convex downward.

(5) If f is a continuous function in the interval $[a , b]$ and for every $X_1 , X_2 \in [a , b]$, then $\grave{f}(X_2) - \grave{f}(X_1) > 0$ when $X_2 > X_1$, then ..... in the interval $]a , b[$

(a) the function f increases. (b) the function f decreases.

(c) the curve of f convex upward. (d) the curve of f convex downward.

(6) If the curve of the function $y = f(X)$ has no inflection point and $\acute{y} = (X-5)^m (X+1)^n$ where $m , n \in \mathbb{Z}^+$, $X \in \mathbb{R}$, then the least value of the expression $(m^2 + n^3)$ is .....

(a) zero (b) 4 (c) 8 (d) 12

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18 Find a , b where the curve :

$X^2 y + a y + b X^2 = 0$ has an inflection point at the point $(1 , -1)$ « 3 , 4 »

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19 *(1st Session 2008)* If $y = f(X)$ represents a polynomial function of third degree such that $\acute{f}(X) < 0$ when $X < \frac{-2}{3}$, $\acute{f}(X) > 0$ when $X > \frac{-2}{3}$ and the curve of the function passes through the point $(1 , 6)$ and there exists a critical point at $(-1 , 2)$ find the equation of this curve and the type of each critical point. « $y = X^3 + 2X^2 + X + 2$ »